

A Finite Overlap Obstruction in the Condition-(3)-Free Core

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Status

This is a partial result and obstruction packet for Argyros–Petsoulas, *A c_0 saturated Banach space with tight structure*, arXiv:1012.2758 [1]. It does not solve whether the source space $\mathfrak{X}_0$ is strictly quasi-prime. It also does not disprove all tight properties of the variant obtained by deleting condition (3) from the definition of G_0 . It proves a narrow obstruction: after condition (3) is removed, bounded norm alone cannot exclude finite overlapping families of pairwise different weights.

Source Passage

The source paper states in the introduction that condition (3) in the definition of G_0 is critical for proving many properties of $\mathfrak{X}_0$, and that the variant omitting condition (3) is still c_0 -saturated but its further properties are not known. The relevant source passage is on page 2 of the PDF:

$\mathfrak{X}_0$ from a standard HI example (for example the Gowers–Maurey space (cf [GM])) is that here we use the odd (i.e. conditional) operations only once at the final step of the definition of W_0 . The familiar reader will also observe that the usual definition of a HI space does not use condition (3) of the definition of G_0 . Condition (3) is critical for proving many of the properties of $\mathfrak{X}_0$ as well as $\mathfrak{L}(\mathfrak{X}_0)$. The particular use of this, is in Proposition 2.1. However it is worth noticing that the variant of the definition not including condition (3) of G_0 also yields a c_0 saturated space and we do not know if this space satisfies the further properties of the space $\mathfrak{X}_0$.

Definitions

Let (m_j) and (n_j) be the sequences from [1]. Let G_0^- be the minimal subset of $c_{00}(\mathbb{N})$ satisfying the first three closure requirements in the definition of G_0 : it contains the coordinate functionals e_t^* , is symmetric, and is closed under the even operations

$$(f_1 < \cdots < f_d, d \leq n_{2j}) \mapsto \frac{1}{m_{2j}} \sum_{i=1}^d f_i.$$

Thus G_0^- is the G_0 -core with the Hilbertian closure under rational ℓ_2 -combinations of pairwise different weights deleted. Let W_0^- be the corresponding once-special extension, defined as in [1] but using G_0^- in place of G_0 .

Proof Intuition

The original proof uses condition (3) exactly when many functionals of pairwise different weights all act on the same block vector. Condition (3) turns such an overlapping family into one G_0 -functional by an ℓ_2 -combination. Once condition (3) is deleted, the remaining even operations only combine successive functionals. The smallest possible overlap, where every functional is supported on the same coordinate, already survives with norm one. Therefore no repair based only on boundedness and different weights can reproduce that step.

Theorem

Theorem 1 (finite overlapping different-weight families). *Fix $N \in \mathbb{N}$, $\eta > 0$, and pairwise distinct even indices*

$$1 \leq j_1 < j_2 < \dots < j_N.$$

There exists an odd level r such that for every coordinate $t \in \mathbb{N}$ the vector $x = e_t$ satisfies

$$\|x\|_{G_0^-} = \|x\|_{W_0^-} = 1$$

and the functionals

$$g_i = \frac{1}{m_{2j_i}} e_t^*, \quad i = 1, \dots, N,$$

belong to G_0^- , have pairwise different weights m_{2j_i} , have the same range $\{t\}$, and satisfy

$$g_i(x) > \frac{\eta}{n_{2r+1}} \quad (i = 1, \dots, N).$$

Proof. Choose r so large that $n_{2r+1} > \eta m_{2j_N}$. For each i , the $d = 1$ case of the even operation applied to e_t^* gives

$$g_i = \frac{1}{m_{2j_i}} e_t^* \in G_0^-,$$

and by the source convention this functional has weight m_{2j_i} . The weights are pairwise different by the choice of the j_i 's, and all the ranges are the singleton $\{t\}$.

Since $e_t^* \in G_0^- \subseteq W_0^-$, both norms of e_t are at least one. Conversely, every functional in W_0^- has ℓ_∞ -norm at most one. This follows by induction from the coordinate functionals: symmetry preserves ℓ_∞ -norm, even averages of successive functionals have ℓ_∞ -norm at most one, the once-special odd averages have ℓ_∞ -norm at most one because their summands are successive, and interval restrictions do not increase ℓ_∞ -norm. Hence $|h(e_t)| \leq 1$ for all $h \in W_0^-$, and

$$\|e_t\|_{G_0^-} = \|e_t\|_{W_0^-} = 1.$$

Finally,

$$g_i(e_t) = \frac{1}{m_{2j_i}} \geq \frac{1}{m_{2j_N}} > \frac{\eta}{n_{2r+1}},$$

as required. □

Corollary 1 (obstruction to the direct repair). *In the condition-(3)-free variant, finite families of pairwise different weights can overlap completely on a single bounded vector. Therefore the case of the source proof that combines such a family by condition (3) cannot be recovered from the remaining even operations plus bounded G_0^- -norm alone.*

Proof. Apply the theorem with arbitrary N . The resulting functionals have common range $\{t\}$, so no two of them are successive. The remaining closure operation in G_0^- combines only successive families. Thus the deleted Hilbertian closure is exactly the rule that allowed the source proof to combine overlapping different-weight functionals. $\square$

Verification Notes and Limitations

The proof is finite and formal; no computational search is used. The only structural facts used are the $d = 1$ case of the even operation and the standard ℓ_∞ -bound for functionals generated by these norming rules.

This packet should be reviewed as a sharp obstruction, not as a full answer to the source paper’s open questions. It does not prove that the analogue of the key averaging proposition fails, nor does it construct a complemented subspace witnessing non-strict-quasi-primeness. It shows that one natural repair route for the condition-(3)-free variant is blocked at the finite combinatorial level.

Bounded novelty check: local run indexes for arXiv:1012.2758 had no promoted solution packet, only a prior attempt note. External searches for the exact source title, “ $\mathfrak{X}_0$ strictly quasi-prime”, and the condition-(3)-free G_0 variant found the source paper and the related Zisimopoulou Bourgain–Delbaen sum paper [2], but no direct resolution of the specific $\mathfrak{X}_0$ question or this finite obstruction.

Human Review Recommendation

Check that the $d = 1$ even operation is admissible under the source definition of weight and that the ℓ_∞ -bound for W_0^- functionals is inherited exactly as stated. The mathematical claim should remain scoped to the finite overlap obstruction.

References

- [1] S. A. Argyros and G. Petsoulas, *A c_0 saturated Banach space with tight structure*, arXiv:1012.2758, 2010.
- [2] D. Zisimopoulou, *Bourgain–Delbaen $\mathcal{L}^\infty$ -sums of Banach spaces*, arXiv:1402.6564, 2014.